

SE2012 – Practical 01 – Java Basics

Refreshing Git

Follow the steps below to access Git through the CLI in Linux, temporarily log in, clone your repository, make changes, commit them, push the changes, and pull the latest changes.

1. Installing Git (if not already installed)

```
sudo apt-get update  
sudo apt-get install git
```

2. Temporarily Logging in to Git

If you are using HTTPS, it is recommended to use a personal access token instead of your GitHub password.

Create a Personal Access Token

1. Go to GitHub Settings → Developer Settings → Personal Access Tokens.
2. Generate a new token with the `repo` scope.

Configure Git with Your Credentials

```
git config --global user.name "Your Name"  
git config --global user.email "your-email@example.com"
```

3. Cloning a Repository

```
git clone <<URI of the Repo>>
```

4. Making Changes

```
cd repository-name  
vim example.java  
or  
vi example.java
```

5. Adding Changes

```
git add .
```

6. Committing Changes

```
git commit -m "Your commit message"
```

7. Pushing Changes

```
git push origin main
```

Java Basics

Question 1: Running the First Java Program

- Create a class named **Welcome.java**
- Use the `Scanner` class
- Prompt for first and last name
- Display: *Welcome to the Second Year <<FirstName>>*
<<LastName>>

Question 2: Student Marks Management

Create a menu-driven program to:

- Add student marks
- Update student marks
- Calculate student average
- Calculate subject average
- Calculate total marks

Question 3: Grade Calculation

- 90 and above: Grade A
 - 80–89: Grade B
 - 70–79: Grade C
 - 60–69: Grade D
 - Below 60: Fail
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Notes – Arrays

1D Arrays

```
int[] array = {1, 2, 3, 4, 5};
```

2D Arrays

```
int[][] matrix = {  
    {1, 2, 3},  
    {4, 5, 6},  
    {7, 8, 9}  
};
```

Example Program

```
public class ArrayExample {  
    public static void main(String[] args) {  
        int[] array = {1, 2, 3, 4, 5};  
        for (int element : array) {  
            System.out.println(element);  
        }  
  
        int[][] matrix = {  
            {1, 2, 3},  
            {4, 5, 6},  
            {7, 8, 9}  
        };  
        for (int[] row : matrix) {  
            for (int element : row) {  
                System.out.println(element);  
            }  
        }  
    }  
}
```